

Comparative Study: 8086 to Pentium 4 -- MPCA Module 6

Definition: The evolution of the Intel x86 Microprocessor Architecture from the 16-bit 8086 to the 7th-generation 32/64-bit Pentium 4 demonstrates radical advancements in bus widths, pipeline depths, caching hierarchies, and parallel instruction execution.

1. Comprehensive Master Comparison Table

Feature / Processor	Intel 8086	Intel 80386DX	Intel Pentium (P5)	Intel Pentium II	Intel Pentium 4
Year Introduced	1978	1985	1993	1997	2000
Data Bus Width	16-bit	32-bit	64-bit	64-bit	64-bit
Address Bus Width	20-bit	32-bit	32-bit	36-bit (PAE)	36-bit (64 GB)
Physical Memory	1 MB	4 GB	4 GB	64 GB	64 GB
Virtual Memory	None	64 TB	64 TB	64 TB	64 TB
Clock Frequency	5 - 10 MHz	16 - 33 MHz	60 - 200 MHz	233 - 450 MHz	1.3 - 3.8 GHz
Pipeline Depth	2-stage (BIU/EU)	3-stage	5-stage (Superscalar)	14-stage (Dynamic)	**20 to 31 stages** (Hyper-Pipeline)
Integer Execution	1 inst / ~4 clocks	1 inst / 2 clocks	**2 inst / clock (U/V)**	3 micro-ops / clock	Out-of-Order Engine
L1 Cache	None	None	16 KB (8K I + 8K D)	32 KB (16K I + 16K D)	8 KB D-Cache + 12K micro-op Tra
L2 Cache	None	External	External	512 KB (Cartridge)	**256 KB - 2 MB On-die**
Key Innovations	Segmented memory	32-bit MMU, Paging	Superscalar, Dual U/V, BT	Dual Independent Bus (DI	NetBurst, Hyper-Threading, Trace

2. Key Architectural Milestones

1. 8086: Introduced 16-bit computing and segmented memory architecture.

2. 80

3. Quick Recall Flow

8086 (16-bit, 1MB) -> 80386 (32-bit, 4GB, Paging) -> Pentium (Superscalar U/V, 64-bit bus) -> Pentium 4 (NetBurst, Hyper-Threading)